The **Sigmoid activation function** is commonly used in neural networks when we want to convert a value into a number between **0 and 1.**

e≈2.718 is a special mathematical constant called **Euler's number.**
It is a constant

Z = In put value

Z	After Sigmoid
-5.50	0.00
-4.60	0.01
-4.00	0.02
-3.40	0.03
-2.80	0.06
-2.20	0.10
-1.60	0.17
-1.00	0.27
-0.40	0.40
0.20	0.55
0.80	0.69
1.40	0.80
2.00	0.88
2.60	0.93
3.20	0.96
3.80	0.98
4.40	0.99
5.00	0.99
6.50	1.00
8.00	1.00

$$SigmoidActivation = \frac{1}{1 + e^{-z}}$$

$$e = 2.718$$

$$= \frac{1}{1 + 2.718^{-(-5.50)}}$$

$$= 0.00$$

$$= \frac{1}{1 + 2.718^{-(-4.60)}}$$

$$= 0.01$$

$$= \frac{1}{1 + 2.718^{-(-4.00)}}$$

$$= 0.02$$